Information Security

Mock 5

Academic Year 2026–2027

Duration: 1 h 30

Instructions. Answer all questions. Responses must be *concise* and every claim must be *justified*. Calculators and external material are not permitted.

Part I – Web Privacy and Data Protection

An online discussion forum is hosted at `forum.example`. Its relevant features are:

- Each registered member has a public profile page, served from `http://forum.example/u/<member>/`, on which the member may place limited custom HTML and JavaScript. For example, a member named Alice controls the page at `http://forum.example/u/alice/`.
- Moderators use a moderation panel at `http://forum.example/mod/`. Members and moderators sign in with a username and password; on success the site issues a session cookie.
- The server accepts connections over both HTTP and HTTPS.

The session cookie has the following attributes:

Attribute	Value
Name	`sid`
Value	(random session identifier)
Path	`/`
Domain	`forum.example`
SameSite	`Lax`

Neither the `Secure` attribute nor the `HttpOnly` attribute is set.

Q1. Bob, a moderator, uses the forum from a public Wi-Fi network on which Eve can both read and alter traffic, and his browser's first contact with `forum.example` is made over HTTP. Explain how Eve can obtain the value of Bob's `sid` cookie.

Q2. Alice may place arbitrary HTML and JavaScript on her profile page at `http://forum.example/u/alice/`, but holds no other access: she cannot alter the forum software and cannot read its logs. Determine whether Alice can obtain the `sid` cookie of a moderator who views her profile. If she can, describe concretely how she would construct her page to capture it; if she cannot, explain what prevents her.

Q3. The forum operators weigh several changes to the way the cookie is issued.

(a) Suppose the `HttpOnly` attribute is added to the cookie. State, with reasons, how this affects your answers to Questions 1 and 2.

(b) Suppose instead that the `Secure` attribute is added (and `HttpOnly` remains unset). State, with reasons, how this affects your answers to Questions 1 and 2.

Q4. For the two independent scenarios below, return to Alice's situation in Question 2 and disregard the attribute changes considered in Question 3.

(a) The moderation panel and the sign-in service are moved to a separate registrable domain, `mod-forum.example`, and the cookie's `Domain` attribute is set to `mod-forum.example`; profile pages remain on `forum.example`. Does your answer to Question 2 still hold? Explain.

(b) Alternatively, the forum keeps the cookie's `Domain` as `forum.example`, but profile pages are moved to per-member subdomains, so that Alice's page is now at `alice.forum.example`. Does your answer to Question 2 still hold? Explain.

Part II – Cryptography

Scenario. Alice is assembling cryptographic building blocks for a project. The adversary Eve can observe all network traffic, store the messages she sees, and on occasion inject or replace messages of her own.

A. Public-key cryptography

Alice decides to use RSA for the project's public-key operations.

Q1. Alice sets up RSA with the two prime numbers $p = 13$ and $q = 11$, and chooses the public exponent $e = 7$.

(a) Compute the modulus n and the value $\varphi(n)$.

(b) Determine the private exponent d. Show the steps of your computation.

(c) Encrypt the message $m = 20$. Show the intermediate steps of the modular exponentiation.

Q2. Bob wants assurance that a document he receives genuinely originates from Alice and has not been altered in transit. Using public-key cryptography together with a cryptographic hash function, describe how Alice can produce such an assurance and how Bob verifies it. Give the sequence of steps.

B. Analysis of a two-party game protocol

Alice and Bob want to play a single round of rock–paper–scissors over an insecure network, with no trusted third party. Each must choose one of *rock*, *paper*, or *scissors*, and the winner is decided by the usual rules. The design goals are that neither party can learn the other's choice before committing to its own, and that neither can change its choice after learning the other's.

Q3. Consider the naive protocol in which each party simply sends its choice, in the clear, to the other. Identify the opportunities for cheating that this protocol permits.

Q4. To address this, Alice proposes the following scheme:

- Each party encrypts its own choice under a secret key it selects (Alice uses K_A, Bob uses K_B) and sends the resulting ciphertext.
- Once both ciphertexts have been exchanged, each party reveals its key, and the other decrypts and learns the choice.

(a) Identify any remaining opportunities for cheating, and justify your answer.

(b) Could the symmetric cipher be replaced by a public-key encryption scheme? Would such a substitution make the protocol secure? Justify your answer.

Q5. Alice and Bob now consider using a cryptographic hash function H.

(a) Design a protocol, based on H, that meets the two design goals stated above. Specify the messages exchanged and the verification each party performs.

(b) Define the *collision-resistance* property of a cryptographic hash function. What happens to your protocol if H fails to be collision-resistant? Explain.

(c) Define the *preimage-resistance* property of a cryptographic hash function. What happens to your protocol if H fails to be preimage-resistant? Explain.

— End of questions —